

A CLASSIFICATION OF METRIZABLE THETA GRAPHS

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ABSTRACT. Cizma and Linial introduced graph metrizability through consistent systems of geodesic paths and asked for a classification of the metrizable theta graphs. They proved that $\Theta_{a,b,c}$ is metrizable when $\min\{a, b, c\} = 1$ and exhibited $\Theta_{3,3,4}$ as a topologically minimal non-metrizable graph. They left open the cases $\min\{a, b, c\} = 2$ and $(a, b, c) = (3, 3, 3)$. We resolve this problem. After sorting $a \leq b \leq c$, the theta graph $\Theta_{a,b,c}$ is metrizable if and only if $a \leq 2$ or $(a, b, c) = (3, 3, 3)$. The infinite family $\Theta_{2,b,c}$ is handled by a direct potential construction based on Ferrers relations. The exceptional graph $\Theta_{3,3,3}$ is handled by an exact finite certificate: all 1587 neighborly consistent path systems are enumerated, and each is supplied with a positive integer edge weighting verified by integer arithmetic.

1. INTRODUCTION

Shortest paths in weighted graphs are usually studied from a metric viewpoint: one begins with positive edge lengths and then studies the induced geodesics. Cizma and Linial proposed the reverse problem in [5]. A *consistent path system* in a graph is a choice of one simple path $P_{u,v}$ between every pair of vertices u, v , with the property that every subpath of a chosen path is itself the chosen path between its endpoints. Such a system is *metrizable* if positive edge lengths can be assigned so that every chosen path is a shortest path. A graph is metrizable if every consistent path system in it is metrizable.

This notion is related to, but distinct from, Bodwin's work on partial systems of unique shortest paths [1]. Bodwin characterizes strict metrizability of partial path systems, whereas graph metrizability asks for all full consistent path systems in a fixed graph, allowing ties among shortest paths. Cizma and Linial proved several foundational results: cycles and outerplanar graphs are strictly metrizable; metrizability is closed downward under topological minors; there is a finite forbidden-topological-minor characterization; and graph metrizability is polynomial-time decidable in principle [5]. Subsequent work constructed irreducible non-metrizable path systems [4] and developed a structural theory of metrizable graphs [2]. The strict version was further separated structurally by the recent result that strictly metrizable graphs are minor-closed [3].

One of the concrete open problems in [5] asks for the metrizability classification of theta graphs. The theta graph $\Theta_{a,b,c}$ consists of two branch vertices s, t joined by three internally disjoint s - t paths of lengths a, b, c . Cizma and Linial observed that $\Theta_{a,b,c}$ is metrizable when $\min\{a, b, c\} = 1$, because such graphs are outerplanar, and they exhibited $\Theta_{3,3,4}$ as a topologically minimal non-metrizable graph. Thus, after sorting $a \leq b \leq c$, the open cases were exactly $a = 2$ and $(a, b, c) = (3, 3, 3)$.

Our main result is the following classification.

Theorem 1.1. *Let $a \leq b \leq c$. Then*

$$\Theta_{a,b,c} \text{ is metrizable} \iff a \leq 2 \text{ or } (a, b, c) = (3, 3, 3).$$

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Equivalently, $\Theta_{a,b,c}$ is non-metrizable if and only if

$$a \geq 3, \quad b \geq 3, \quad c \geq 4.$$

The non-metrizable direction follows from the already known obstruction $\Theta_{3,3,4}$ and topological-minor closure. The main theoretical part of the paper proves that every $\Theta_{2,b,c}$ is metrizable. The proof turns consistency into two types of Ferrers relations and then realizes these relations by one-dimensional potentials. The only remaining positive case, $\Theta_{3,3,3}$, is handled by a finite exact certificate. The certificate is not numerical: for each of the 1587 neighborly consistent path systems, it records a positive integer edge weighting, and a verifier checks all shortest-path inequalities by integer arithmetic.

2. PRELIMINARIES

All graphs are finite and connected, and all paths are simple.

Definition 2.1. Let $G = (V, E)$ be a graph. A *path system* $\mathcal{P}$ in G is a collection

$$\mathcal{P} = \{P_{u,v} : \{u, v\} \in \binom{V}{2}\},$$

where $P_{u,v}$ is a simple u - v path. It is *consistent* if, whenever x, y lie on $P_{u,v}$, the x - y subpath of $P_{u,v}$ is $P_{x,y}$. It is *neighborly* if $P_{u,v} = uv$ for every edge $uv \in E$.

Definition 2.2. A positive edge weighting $w : E \rightarrow (0, \infty)$ induces a path system $\mathcal{P}$ if every $P_{u,v}$ is a w -shortest u - v path. A path system is *metrizable* if it is induced by some positive edge weighting. A graph is *metrizable* if every consistent path system in it is metrizable.

We use the following facts from [5].

Proposition 2.3. *The following hold.*

- (i) *Cycles are strictly metrizable.*
- (ii) *Outerplanar graphs are strictly metrizable.*
- (iii) *Metrizability is closed downward under topological minors. Consequently, if a graph contains a subdivision of a non-metrizable graph, then it is non-metrizable.*

We also use the cycle structure theorem of Cizma and Linial: every consistent path system on a cycle becomes the standard geodesic system on an odd cycle after contracting its persistent edges [5, Proposition 4.8].

3. FIRST REDUCTIONS FOR THETA GRAPHS

Proposition 3.1. *Let $G = \Theta_{a,b,c}$. Every non-neighborly consistent path system on G is metrizable.*

Proof. Let $\mathcal{P}$ be non-neighborly. Choose an edge $e = xy$ for which $P_{x,y} \neq xy$. No path of $\mathcal{P}$ can use e : if e appeared in some $P_{u,v}$, then the x - y subpath of $P_{u,v}$ would be the edge xy , and consistency would force $P_{x,y} = xy$.

Thus all paths of $\mathcal{P}$ lie in $G - e$. The graph $G - e$ is outerplanar, so by Proposition 2.3 it has a positive edge weighting w_0 inducing $\mathcal{P}$ as a path system on $G - e$. Extend w_0 to G by assigning e a weight larger than the total w_0 -weight of all edges in $G - e$. Then no path using e can be shorter than a prescribed path, since every prescribed path avoids e . Hence the extended weighting induces $\mathcal{P}$ on G . $\square$

Corollary 3.2. *Any non-metrizable path system on a theta graph is neighborly.*

Proposition 3.3. *Let $a \leq b \leq c$. If $a \geq 3$, $b \geq 3$, and $c \geq 4$, then $\Theta_{a,b,c}$ is non-metrizable.*

Proof. The graph $\Theta_{a,b,c}$ contains a subdivision of $\Theta_{3,3,4}$. The graph $\Theta_{3,3,4}$ is non-metrizable by the certificate displayed as Fig. 22k in [5]. The conclusion follows from topological-minor closure, Proposition 2.3(iii). $\square$

For reference, we record the local algebraic form of the obstruction.

Proposition 3.4 (The six-inequality block). *Suppose a graph contains internally disjoint sub-paths*

$$A : s - a_1 - a_2 - t, \quad B : s - b_1 - b_2 - t, \quad C : s - c_1 - c_2 - c_3 - t.$$

If a path system contains the six prescribed paths

$$\begin{aligned} P_{a_1,b_2} &= a_1 s b_1 b_2, & P_{a_2,b_1} &= a_2 t b_2 b_1, \\ P_{a_1,c_3} &= a_1 a_2 t c_3, & P_{a_2,c_2} &= a_2 a_1 s c_1 c_2, \\ P_{b_1,c_3} &= b_1 s c_1 c_2 c_3, & P_{b_2,c_1} &= b_2 t c_3 c_2 c_1, \end{aligned}$$

then no positive edge weighting can induce all six paths.

Proof. Write

$$\begin{aligned} A_1 &= w(sa_1), & A_2 &= w(a_1a_2), & A_3 &= w(a_2t), \\ B_1 &= w(sb_1), & B_2 &= w(b_1b_2), & B_3 &= w(b_2t), \\ C_1 &= w(sc_1), & C_2 &= w(c_1c_2), & C_3 &= w(c_2c_3), & C_4 &= w(c_3t). \end{aligned}$$

If the six displayed paths are geodesics, then comparison with the natural alternative on the same endpoints gives

$$\begin{aligned} A_1 + B_1 + B_2 &\leq A_2 + A_3 + B_3, \\ A_3 + B_3 + B_2 &\leq A_2 + A_1 + B_1, \\ A_2 + A_3 + C_4 &\leq A_1 + C_1 + C_2 + C_3, \\ A_2 + A_1 + C_1 + C_2 &\leq A_3 + C_4 + C_3, \\ B_1 + C_1 + C_2 + C_3 &\leq B_2 + B_3 + C_4, \\ B_3 + C_4 + C_3 + C_2 &\leq B_2 + B_1 + C_1. \end{aligned}$$

Adding these inequalities and cancelling common terms yields $2C_2 \leq 0$, impossible for positive edge weights. $\square$

4. A CYCLE-CUT LEMMA

The next lemma is used to handle the case $\Theta_{2,b,c}$ in which the length-two arm is not the chosen s - t path.

Lemma 4.1 (Cycle-cut realization). *Let H be a cycle with two distinct vertices s, t . Let $\mathcal{Q}$ be a consistent path system on H . Suppose that a new vertex a is joined to s and t , and that a consistent extension prescribes each a - v path, $v \in V(H)$, to begin either with as or with at . Then there is a positive edge weighting w_H inducing $\mathcal{Q}$ and a real number h such that*

$$\begin{aligned} P_{a,v} \text{ begins with } as &\implies d_{w_H}(s, v) - d_{w_H}(t, v) < h, \\ P_{a,v} \text{ begins with } at &\implies d_{w_H}(s, v) - d_{w_H}(t, v) > h. \end{aligned}$$

Proof. By the cycle structure theorem [5, Proposition 4.8], after contracting persistent edges the system $\mathcal{Q}$ becomes the standard geodesic system on an odd cycle C_{2n+1} . Persistent edges may later be expanded with sufficiently small positive weights, so it suffices to handle the reduced case.

Label the odd cycle by $0, 1, \dots, 2n$ in cyclic order, put $s = 0$, and write $t = d$ with $1 \leq d \leq n$. Consistency of the extension implies that the vertices whose $a-v$ path begins with as form a prefix on each of the two $s-t$ arcs. Thus, for some $0 \leq q < d$ and $n - d \leq r \leq n$, this set is

$$S = \{2n + 1 - r, \dots, 2n, 0, 1, \dots, q\}.$$

The inequalities on q, r express that standard shortest paths from s to vertices of S remain in S , and standard shortest paths from t to vertices outside S remain outside S .

Let $N = 2n + 1$, and write e_i for the edge joining i to $i + 1$ modulo N . For each k , let R_k put weight 1 on e_k and on e_{k+n} , and weight 0 elsewhere. Every R_k induces the standard odd-cycle path system non-strictly: every standard path contains at most one of the two weighted edges, while the complementary path contains at least one. Let $p = N - r - 1$ and set

$$w_0 = R_q + R_p + \frac{1}{2}R_0.$$

Let $\phi(v) = d_{w_0}(0, v) - d_{w_0}(d, v)$. It remains only to check the two boundary gaps of the cut. Put

$$F_1 = \phi(p) - \phi(q), \quad F_2 = \phi(q + 1) - \phi(p + 1).$$

For the elementary weights one obtains

$$F_1(R_q + R_p) = 2, \quad F_2(R_q + R_p) \in \{0, 2, 4\},$$

and

$$F_1(R_0) \in \{-2, 0, 2\}, \quad F_2(R_0) = 2.$$

Therefore $F_1(w_0) > 0$ and $F_2(w_0) > 0$, i.e.

$$\phi(q) < \phi(p), \quad \phi(p + 1) < \phi(q + 1).$$

Since ϕ is monotone on each $s-t$ arc, these two inequalities imply

$$\max_{v \in S} \phi(v) < \min_{v \notin S} \phi(v).$$

Choose h strictly between these two values.

Finally add a sufficiently small positive multiple of the uniform edge weighting. The standard odd-cycle system becomes strictly induced, and the strict cut separation remains true. Expanding persistent edges with sufficiently small positive weights preserves strict induction and the strict cut separation. This proves the lemma. $\square$

Proposition 4.2. *Let $G = \Theta_{2,b,c}$, and let its length-two arm be $A = s - a - t$. If $\mathcal{P}$ is neighborly and $P_{s,t} \neq A$, then $\mathcal{P}$ is metrizable.*

Proof. Let $H = G - a$. Then H is the cycle formed by the two long arms. No prescribed path whose endpoints lie in H can pass through a , because such a path would contain A as its $s-t$ subpath, forcing $P_{s,t} = A$ by consistency. Therefore $\mathcal{P}$ restricts to a cycle path system $\mathcal{Q}$ on H .

Apply Lemma 4.1 to obtain w_H and h . Choose $M > |h|$ so large that no shortest path between two vertices of H can improve by going through a , and set

$$w(as) = M, \quad w(at) = M + h.$$

Both weights are positive. For $v \in V(H)$, the route from a through s is no longer than the route through t exactly when

$$d_{w_H}(s, v) - d_{w_H}(t, v) \leq h.$$

Thus every prescribed $a-v$ path is a geodesic, and the choice of M preserves all prescribed paths inside H . $\square$

5. FERRERS REPRESENTATIONS

We use two elementary forms of the same fact: Ferrers relations on chains are threshold relations.

Lemma 5.1 (Additive Ferrers representation). *Let $I = \{1, \dots, m\}$ and $J = \{1, \dots, n\}$ be chains. Let $F \subseteq I \times J$ satisfy*

$$(i, j) \in F, \quad i' \leq i, \quad j' \leq j \implies (i', j') \in F.$$

Then there are strictly increasing real numbers x_i and y_j such that

$$(i, j) \in F \iff x_i + y_j < 0.$$

Moreover, if a prefix cut is prescribed in I and a prefix cut is prescribed in J , the representation may be chosen so that both cuts are represented by a common threshold h :

$$i \text{ lies in the prescribed prefix of } I \iff x_i < h,$$

$$j \text{ lies in the prescribed prefix of } J \iff y_j < h.$$

Proof. For each i , let r_i be the largest j such that $(i, j) \in F$, with $r_i = 0$ if the i th row is empty. The sequence r_i is non-increasing. Put $x_i = i$. For $j \in J$, let

$$k_j = \max\{i : r_i \geq j\},$$

with $k_j = 0$ if the set is empty. Choose a sufficiently small $\delta > 0$ and set

$$y_j = -k_j - \frac{1}{2} + \delta j.$$

The values y_j are strictly increasing because k_j is non-increasing in j , and for δ small enough the inequalities $x_i + y_j < 0$ hold exactly when $i \leq k_j$, equivalently when $j \leq r_i$.

Now suppose prefix cuts are prescribed. Starting with any representation, replace

$$x_i \mapsto \lambda x_i + \alpha, \quad y_j \mapsto \lambda y_j - \alpha,$$

where $\lambda > 0$. This preserves all signs of $x_i + y_j$. The interval of possible thresholds for the I -cut moves to the right as α increases, while the interval of possible thresholds for the J -cut moves to the left. Hence, for a suitable α , the two intervals overlap. Choose h in their intersection. Empty and full prefixes are handled by placing h outside the corresponding range. $\square$

Lemma 5.2 (Difference Ferrers representation). *Let $D : s = d_0, d_1, \dots, d_m = t$ be a chain. Let D^- be a prefix containing s , let D^+ be a suffix containing t , and let $D^0 = D \setminus (D^- \cup D^+)$. Suppose $F \subseteq D^- \times D^+$ satisfies*

$$(d_i, d_j) \in F, \quad i' \leq i, \quad j' \geq j \implies (d_{i'}, d_{j'}) \in F,$$

and assume that the row s and the column t belong entirely to F . Fix $L > 1$. Given any strictly increasing prescribed potentials in $(-1, 1)$ on D^0 , there are $M > L$ and a strictly increasing sequence

$$-M = X_0 < X_1 < \dots < X_m = M$$

which extends those middle potentials, has $X_i < -L$ on $D^- \setminus \{s\}$ and $X_i > L$ on $D^+ \setminus \{t\}$, and satisfies

$$(d_i, d_j) \in F \iff X_j - X_i > M + L$$

for all $d_i \in D^-$ and $d_j \in D^+$ with $i < j$.

Proof. Reverse the order on D^+ . The displayed hypothesis becomes an ordinary Ferrers condition. By Lemma 5.1, after an affine rescaling there are strictly increasing numbers U_i on D^- and strictly increasing numbers V_j in the reversed order on D^+ , and a threshold $R > 0$, such that

$$(d_i, d_j) \in F \iff U_i + V_j < R.$$

We may shift the two families by opposite constants so that $U_s = 0$ and $V_t = 0$. Since the row s and column t are entirely in F , all relevant U_i and V_j are strictly smaller than R . Enlarging the scale if necessary, assume R is large enough that the prescribed middle potentials lie in the gap between the prefix values below $-L$ and the suffix values above L .

Set $M = R + L$, and define

$$X_i = -M + U_i \quad (d_i \in D^-), \quad X_j = M - V_j \quad (d_j \in D^+).$$

Since V_j is increasing in the reversed order, the values $M - V_j$ are strictly increasing in the natural order along D^+ . Also $X_s = -M$, $X_t = M$, all non-end prefix values are below $-L$, and all non-end suffix values are above L . Finally,

$$X_j - X_i = 2M - (U_i + V_j),$$

so

$$X_j - X_i > M + L \iff U_i + V_j < R.$$

Together with the prescribed middle potentials, this gives the required strictly increasing sequence. $\square$

6. THE FAMILY $\Theta_{2,b,c}$

Theorem 6.1. *For every $2 \leq b \leq c$, the theta graph $\Theta_{2,b,c}$ is metrizable.*

Proof. Let the length-two arm be

$$A = s - a - t,$$

and write the other arms as

$$B : s = b_0, b_1, \dots, b_b = t, \quad C : s = c_0, c_1, \dots, c_c = t.$$

Let $\mathcal{P}$ be a consistent path system. By Proposition 3.1, we may assume $\mathcal{P}$ is neighborly. By Proposition 4.2, we may assume

$$P_{s,t} = A = s - a - t.$$

For a long arm $D : s = d_0, d_1, \dots, d_m = t$, define

$$D^- = \{s\} \cup \{d_i : 1 \leq i \leq m-1, P_{d_i,t} = d_i D \text{sat}\},$$

$$D^+ = \{t\} \cup \{d_i : 1 \leq i \leq m-1, P_{s,d_i} = \text{sat} D d_i\},$$

and set $D^0 = D \setminus (D^- \cup D^+)$. Consistency implies that D^- is a prefix, D^+ is a suffix, and $D^- \cap D^+ = \emptyset$. For example, if $d_i \in D^-$ and $i' \leq i$, then the $d_{i'}-t$ subpath of $P_{d_i,t}$ is $d_{i'} D \text{sat}$, so $d_{i'} \in D^-$. The assertion for D^+ is symmetric. If an internal vertex d_i belonged to both D^- and D^+ , then the $s-d_i$ subpath of $P_{d_i,t}$ would be the direct subpath $s D d_i$, whereas P_{s,d_i} would be $\text{sat} D d_i$, contradicting consistency.

For $i < j$ on the same long arm D , if P_{d_i,d_j} uses A , then $d_i \in D^-$ and $d_j \in D^+$. The set of such pairs is Ferrers: if (d_i, d_j) uses A , $i' \leq i$, and $j' \geq j$, then P_{d_i,d_j} contains $d_{i'} D \text{sat} D d_{j'}$ as a subpath, so $P_{d_{i'},d_{j'}}$ also uses A .

Next consider $b_i \in B$ and $c_j \in C$. The possible routes are

$$S : b_i B s C c_j, \quad T : b_i B t C c_j, \quad U : b_i B \text{sat} C c_j, \quad V : b_i B t a s C c_j.$$

Consistency gives

$$\begin{aligned} P_{b_i, c_j} = U &\iff b_i \in B^- \text{ and } c_j \in C^+, \\ P_{b_i, c_j} = V &\iff b_i \in B^+ \text{ and } c_j \in C^-. \end{aligned}$$

Indeed, the forward implication follows by taking the b_i-t and $s-c_j$ subpaths. Conversely, if $b_i \in B^-$ and $c_j \in C^+$, then the routes S, T, V would give a wrong b_i-t or $s-c_j$ subpath, so only U is possible. The proof for V is symmetric.

Outside the two rectangles $B^- \times C^+$ and $B^+ \times C^-$, the route is either S or T . More precisely, away from the two exceptional rectangles, membership in B^- or C^- forces the route through s , while membership in B^+ or C^+ forces the route through t ; otherwise one obtains a wrong b_i-t , $s-c_j$, $s-b_i$, or c_j-t subpath. On the central rectangle $B^0 \times C^0$, the S -region is Ferrers: if $P_{b_i, c_j} = S$, $i' \leq i$, and $j' \leq j$, then $P_{b_{i'}, c_{j'}}$ contains $b_{i'}BsCc_{j'}$ as a subpath, hence $P_{b_{i'}, c_{j'}} = S$.

Fix $L = 2$. The vertices d_i of any long arm whose $a-d_i$ path begins with as form a prefix: if $i' \leq i$ and $P_{a, d_{i'}}$ begins with as , then the $a-d_{i'}$ subpath of P_{a, d_i} also begins with as . If one of B^0, C^0 is empty, choose arbitrary small increasing middle potentials realizing these prefix cuts. Otherwise apply Lemma 5.1 to the Ferrers S -region on $B^0 \times C^0$, together with the prefix cuts on B^0 and C^0 determined by whether P_{a, b_i} and P_{a, c_j} begin with as . This gives strictly increasing middle potentials X_i for $b_i \in B^0$, Y_j for $c_j \in C^0$, and a threshold h , such that

$$P_{a, b_i} \text{ begins with } as \iff X_i < h, \quad P_{a, c_j} \text{ begins with } as \iff Y_j < h,$$

and

$$P_{b_i, c_j} = S \iff X_i + Y_j < 0 \quad (b_i \in B^0, c_j \in C^0).$$

Scaling all X_i, Y_j, h by a small positive factor, assume $|X_i|, |Y_j|, |h| < 1$.

Now apply Lemma 5.2 separately to the same-arm Ferrers relations on $B^- \times B^+$ and $C^- \times C^+$, preserving the already chosen middle potentials. We obtain strictly increasing sequences

$$\begin{aligned} -M_B = X_0 < X_1 < \dots < X_b = M_B, \\ -M_C = Y_0 < Y_1 < \dots < Y_c = M_C, \end{aligned}$$

with prefix values below $-L$, middle values in $(-1, 1)$, suffix values above L , and with same-arm paths through A represented exactly by

$$X_j - X_i > M_B + L \quad \text{or} \quad Y_j - Y_i > M_C + L.$$

Define edge weights by

$$\begin{aligned} w(sa) &= \frac{L-h}{2}, & w(at) &= \frac{L+h}{2}, \\ w(b_{i-1}b_i) &= \frac{X_i - X_{i-1}}{2}, & w(c_{j-1}c_j) &= \frac{Y_j - Y_{j-1}}{2}. \end{aligned}$$

All weights are positive. Moreover,

$$d_B(s, b_i) - d_B(t, b_i) = X_i, \quad d_C(s, c_j) - d_C(t, c_j) = Y_j.$$

We verify that every prescribed path is a geodesic. For a and a long-arm vertex v , the route through s is no longer than the route through t if and only if

$$d(s, v) - d(t, v) \leq h,$$

which is exactly how the threshold was chosen.

For two vertices d_i, d_j on the same long arm D , $i < j$, let Z denote the corresponding potential sequence and let M_D be the total length of the arm. The direct route along D has length $(Z_j - Z_i)/2$, while the route through A has length

$$M_D + L - \frac{Z_j - Z_i}{2}.$$

Thus the comparison is exactly $Z_j - Z_i \leq M_D + L$ versus $Z_j - Z_i \geq M_D + L$, which is realized by Lemma 5.2. The route through the other long arm is at least as long as the route through A , because the other long arm has total length larger than L .

Finally take $b_i \in B$ and $c_j \in C$. Direct comparison of the four possible routes gives the following criteria for the named route to be geodesic:

$$\begin{aligned} S: & X_i \leq L, \quad Y_j \leq L, \quad X_i + Y_j \leq 0, \\ T: & X_i \geq -L, \quad Y_j \geq -L, \quad X_i + Y_j \geq 0, \\ U: & X_i \leq -L, \quad Y_j \geq L, \\ V: & X_i \geq L, \quad Y_j \leq -L. \end{aligned}$$

The rectangles $B^- \times C^+$ and $B^+ \times C^-$ are exactly the U and V regions. The remaining non-middle regions are forced to be S or T by consistency and by the signs of X_i and Y_j . On $B^0 \times C^0$, the sign of $X_i + Y_j$ gives exactly the prescribed Ferrers S/T matrix. Hence every cross-arm prescribed path is a geodesic. This completes the proof. $\square$

7. THE FINITE GRAPH $\Theta_{3,3,3}$

The graph $\Theta_{3,3,3}$ is small enough for a complete exact certificate. We describe the certificate because the proof is computer-assisted but not numerical in the final verification.

Proposition 7.1. *The graph $\Theta_{3,3,3}$ is metrizable.*

Proof. By Proposition 3.1, it suffices to consider neighborly path systems. The graph $\Theta_{3,3,3}$ has eight vertices and nine edges. A complete enumeration by subpath closure gives exactly 1587 neighborly consistent path systems.

For each such system $\mathcal{P}$, the certificate file records a positive integer vector $w \in \mathbb{Z}_{>0}^E$. The verifier checks, using integer arithmetic, that for every unordered pair u, v and every simple u - v path Q ,

$$w(P_{u,v}) \leq w(Q).$$

Thus the recorded vector induces $\mathcal{P}$. The maximum integer edge weight appearing in the certificates is 5. Therefore every neighborly consistent path system on $\Theta_{3,3,3}$ is metrizable, and the non-neighborly systems are metrizable by Proposition 3.1. $\square$

Remark 7.2 (Reproducibility). The supplementary material contains a self-contained verification script and a compressed JSONL certificate file. The script enumerates all neighborly consistent path systems on $\Theta_{3,3,3}$, verifies that the certificate file contains exactly these systems, and checks the recorded positive integer edge weights against all simple alternative paths. The SHA256 hash of the compressed certificate file supplied with this submission is

$$9606a50e78eaa001d92e1f68a9b0263a96a581f650e8ee352e3015d2fc362bbc.$$

The SHA256 hash of the verification script is

$$a79af295b41ab319e3b12aadd3aee10adce935f451db958848e6f36eb5875db.$$

8. PROOF OF THE CLASSIFICATION

Proof of Theorem 1.1. If $a = 1$, then $\Theta_{a,b,c}$ is outerplanar, hence metrizable by Proposition 2.3. If $a = 2$, metrizability follows from Theorem 6.1. The remaining positive case $(a, b, c) = (3, 3, 3)$ is Proposition 7.1.

Conversely, if $a \geq 3$, $b \geq 3$, and $c \geq 4$, then non-metrizability follows from Proposition 3.3. These alternatives are exhaustive when $a \leq b \leq c$. $\square$

9. CONCLUDING REMARKS

The classification shows that, among theta graphs, the first obstruction is exactly the six-inequality block realized by $\Theta_{3,3,4}$. All longer theta graphs with three arms of lengths at least 3, 3, 4 inherit this obstruction as a topological minor. On the positive side, a length-two arm suppresses the possible obstruction: the proof for $\Theta_{2,b,c}$ turns consistency into Ferrers relations and realizes them by one-dimensional potentials.

The proof of $\Theta_{3,3,3}$ is finite and exact, but computer-assisted. It would be interesting to replace this certificate by a short hand proof, or to extend the potential method to cover this final exceptional graph directly.

DATA AND CODE AVAILABILITY

The supplementary material consists of two files: `jgt_theta333_verify_certificates.py` and `jgt_theta333_exact_integer_certificates.jsonl.gz`. The verifier uses only the Python standard library. It checks the enumeration, consistency, neighborliness, integer positivity, and all shortest-path inequalities for the finite certificate used in Proposition 7.1.

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